

SWEEP rule on the KD-Toric code

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1 Explantation of a Spanned Slice

[COMMENT] Here we follow the original proof, but change the triminalogy, m will be the number of leaves in the explanation, then in the end we conclude that the number of faults is at least m/Δ . So $|R| \leq f(\frac{m}{\Delta})$.

Definition 1.1. Consider a time graph (G, T) , an embedding $f: G \rightarrow \mathbb{R}^d$, such that.. Let $\beta = \frac{d+1}{\xi}$ and $\alpha = \beta \mathbf{Size}(\text{fork})$. Given sets W of points, R of arrows and F of forks, we define U as the set of vertices of the graph induced by the edges of $R \cup F$. The sets W, R and F form an explanation of a spanned slice $\hat{K} = \langle K, v_1, \dots, v_{d+1} \rangle$ if and only if:

1. $W \cup \{v_1, \dots, v_{d+1}\} \subset U \subset H_K$ and the graph $(U, R \cup F)$ is connected.
2. All elements of W are leaves.
3. For $m = |W|$ and $s = \mathbf{Size}(K)$ we have $|R| \leq \alpha(m - 1) - s$ and $|F| < m - 1$.

Claim 1.2. Every spanned slice $K = \langle K, v_1, \dots, v_{d+1} \rangle$ has an explanation.

Proof. Let $h = T(K) - \min_{u \in H_K} T(u)$. The proof is by induction on h .

1. Base: $h = 0$, namely T is constant on H_K , so no arrows connect points of H_K . Thus, H_K , as well as $\mathbf{K}$, consists only of leaves. In that case, $U = W = \{u\}$, $m = 1$, $s = 0$, and $|R|, |F| = 0$ satisfy the requirement.
2. Induction Step: $h \neq 0$ implies $K \neq H_K$. Thus, by ??, we know that for some k there exist spanned slices $\hat{K}_0, \dots, \hat{K}_k$ and forks $F_1, \dots, F_k$ that satisfy ??.

Consider $i \in 0, \dots, k$, denote by $s_i = \mathbf{Size}(\hat{K}_i)$. By the inductive hypothesis and since $T(K_i) = T(K) - 1$, we know that $\hat{K}_i$ has an explanation formed by a set of leaves $\mathbf{W}_i$, a set of arrows $\mathbf{R}_i$ and a set of forks $\mathbf{F}_i$. Let $m_i = |\mathbf{W}_i|$, So the sets R_i and F_i have at most $\alpha m_i - s_i$ and $m_i - 1$ leaves. We define:

- (a) $\mathbf{W} = \cup_{i=0}^k \mathbf{W}_i$, Notice that $\mathbf{W}_i \subset K_i$ and that $T(K_i) = T(K_j)$ for all i, j , thus by ?? $\mathbf{W}$ is a union of disjoint set, Hence W contains $m = \sum_{i=0}^k m_i$ leaves.

- (b) $\mathbf{F} = \{F_1, \dots, F_k\} \cup_{i=0}^k \mathbf{F}_i$ with at most $k + \sum_{i=0}^k m_i - 1 = m - 1$ elements.
(c) $\mathbf{R} = \{\{v_i, \text{pre}_i(v_i)\}\} \cup_{i=0}^k \mathbf{R}_i$ with at most

$$\begin{aligned}
& d + 1 + \sum_{i=0}^k \alpha(m_i - 1) - \beta s_i \\
& \leq d + 1 + \alpha m - \alpha(k + 1) - \beta(s + \xi - k \cdot \mathbf{Size}(\text{fork})) \\
& = \alpha(m - 1) - \beta s + (\alpha + d + 1 - (k + 1)\alpha - \beta\xi + \beta k \cdot \mathbf{Size}(\text{fork}))
\end{aligned}$$

So its enough to show that for $k \geq 0$ the term in the closure is negative, We do that by showing that the slop is negative, and then the value of the term at zero is negative:

$$\begin{aligned}
-\alpha + \beta \mathbf{Size}(\text{fork}) &\leq 0 \Rightarrow \mathbf{Size}(\text{fork}) \leq \frac{\alpha}{\beta} \\
d + 1 - \beta\xi &\leq 0 \Rightarrow \frac{d + 1}{\beta} \leq \xi
\end{aligned}$$

And those inequalities hold for $\beta = \frac{d+1}{\xi}$ and $\alpha = \beta \mathbf{Size}(\text{fork})$. Finally, by ??, the graphs induced by $\mathbf{R}_i \cup \mathbf{F}_i$ are connected, thus the graph induced by $\mathbf{R} \cup \mathbf{F}$ is also connected. Therefore, $\mathbf{W}, \mathbf{R}, \mathbf{F}$ form an explanation of K .

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